지하수의 비소 오염 검출을 위한 저비용 머신러닝 기반 접근법

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A Low-Cost Machine Learning Approach for Arsenic Contamination Detection in Rural Groundwater

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요 약

방글라데시 농촌 지역의 지하수 비소 오염은 수백만 명이 의존하는 얕은 관정이 국가 안전 기준(50 μg/L)을 초과하는 심각한 공중 보건 문제이다. 기존의 실험실 분석 및 상업용 현장 키트는 비용과 시간 문제로 일상적인 지역사회 수준의 스크리닝에 적합하지 않다. 본 연구는 IoT 센서(pH, 전기전도도, 수온)와 간단한 현장 측정(철, 망간) 및 우물 깊이를 결합한 저비용 접근법으로 실험실 없이 안전하지 않은 관정을 식별할 수 있는지 조사한다. BGS/DFID/DPHE 특별 연구 데이터셋(n=271)을 사용하여 로지스틱 회귀, 랜덤 포레스트, 다층 퍼셉트론, XGBoost, 그래디언트 부스팅 등 5가지 분류기를 SMOTE 기법과 함께 훈련 및 평가하였다. 랜덤 포레스트가 AUC=0.923, F1=0.838, 불안전 클래스 재현율=0.94로 최고 성능을 달성하였으며, 전체 실험실 분석 대비 93.8%의 판별 능력을 유지하였다. SHAP 분석을 통해 철 농도, pH, 우물 깊이가 주요 예측 변수임을 확인하였다.

Abstract

Arsenic contamination of groundwater in rural Bangladesh is a serious public health issue, with shallow wells relying on millions of people exceeding the national safety standard (50 μg/L). Existing laboratory analyses and commercial field kits are unsuitable for routine community-level screening due to cost and time constraints. This study investigates whether unsafe wells can be identified without a laboratory using a low-cost approach that combines IoT sensors (pH, electrical conductivity, water temperature) with simple field measurements (iron, manganese) and well depth. Using the BGS/DFID/DPHE Special Study dataset (n=271), five classifiers—logistic regression, random forest, multilayer perceptron, XGBoost, and gradient boosting—were trained and evaluated in conjunction with the SMOTE technique. Random forest achieved the best performance with an AUC of 0.923, F1 of 0.838, and unsafe class recall of 0.94, maintaining a discriminative ability of 93.8% compared to all laboratory analyses. SHAP analysis confirmed that iron concentration, pH, and well depth are key predictor variables.

Key words

Arsenic prediction; Bangladesh groundwater; machine learning; Random Forest; SMOTE; IoT; water quality.

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Ⅰ. Introduction

Safe drinking water is a fundamental human right, yet arsenic contamination in groundwater remains one of the most critical public health challenges in South Asia. Bangladesh faces what the WHO has described as the largest mass poisoning of a population in history [1]. Approximately 20 million people are exposed to arsenic exceeding the WHO guideline of 10 μg/L and Bangladesh's national standard of 50 μg/L through shallow tubewell water [2]. Prolonged exposure causes skin lesions, keratosis, and cancers of the skin, bladder, and lung, and over 55% of tubewells at primary schools exceed the WHO guideline, posing direct risks to children [3] [4].

Conventional arsenic testing relies on laboratory-based AAS and ICP-MS techniques, which are expensive, slow, and unavailable in remote rural areas [5]. This paper addresses these gaps with the following contributions:

1. Five machine learning classifiers are systematically compared for arsenic safety prediction using IoT-feasible input features.
2. The performance trade-off between full laboratory analysis and a minimal IoT sensor set is quantified.
3. SMOTE-based oversampling is applied to handle class imbalance.
4. SHAP values provide interpretable explanations for model predictions.
5. Cross-dataset generalization from a detailed study dataset to a large national survey is validated.

Ⅱ. Background and related work

A. Arsenic Geochemistry in Bangladesh

Arsenic in Bangladesh's groundwater is primarily of geogenic origin [6], mobilized through reductive dissolution of iron oxyhydroxide minerals in Holocene alluvial sediments. When dissolved oxygen is depleted in shallow aquifers, iron-bound arsenic is released into groundwater, explaining why arsenic co-occurs strongly with Fe and Mn and why pH directly influences arsenic mobility [7].

B. Existing Testing Approaches and Their Limitations

The most widely used field method in Bangladesh is the Merck or Hach arsenic field kit, which uses a colorimetric reaction to estimate arsenic concentration.

C. Machine Learning for Water Quality Prediction

Random Forest and gradient boosting methods have demonstrated strong performance in arsenic prediction tasks using hydrochemical datasets [8]. However, most existing studies use full laboratory datasets and do not address the practical constraint of deploying models with only low-cost measurable inputs. SHAP-based explainability has been advocated for environmental ML applications where stakeholder trust is critical.

Ⅲ. Dataset and study area

A. Primary Dataset: BGS/ DFID/ DPHE Special Study

The primary dataset is the BGS/DFID/DPHE SpecialStudy, a detailed hydrochemical survey of groundwater wells across Bangladesh. After cleaning, the working dataset comprises 271 well samples including physical parameters (pH, SEC, temperature, well depth), redox indicators (DO,Eh), major ions (Fe, Mn, Ca, Mg, Na, K, Cl, SO4, HCO3), and total arsenic (AsTot, μg/L). Class distribution at the 50 μg/L threshold:

- Safe (< 50 μg/L): 163 wells (60.1%)
- Unsafe (≥ 50 μg/L): 108 wells (39.9%)

B. Validation Dataset: BGS National Survey

For cross-dataset generalization, the BGS/DFID/DPHE National Survey [9] comprising 3,534 well samples is used. This dataset contains Fe, Mn, and well depth but lacks pH, EC, and temperature, making it suitable for testing generalization to a leaner measurement context.

C. IoT Feature Selection

Six features were selected as the "IoT + Fe/Mn Kit" scenario (Table 1). Total hardware cost: ~USD 20. Per-test consumable cost:~USD 0.50.

Table 1: IoT feature set with measurement method and approximate cost.

IV. Methodology

Feature	Method	Cost($)
pH	Electrochemical sensor	3-5
EC/SEC	Conductivity probe	3-5
Temperature	DS18B20 thermistor	1
Well Depth	Manual/pressue sensor	0-5
Iron(Fe)	Colorimetric strip	0.30/test
Manganese(Mn)	Colorimetric strip	0.20/test

A. Data Preprocessing

Censored values (e.g., <0.04 mg/L) were replaced with half the detection limit (LOD/2) a standard approach in environmental geochemistry.

B. Class Imbalance Handling: SMOTE

The dataset exhibits moderate class imbalance (60:40 atthe 50 µg/L threshold). The Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set, generating synthetic samples by interpolating between real minority-class, using k = 5 nearest neighbors. All SMOTE operations were applied exclusively to training data to prevent leakage.

C. Models

Five classifiers were trained and compared:

1. Logistic Regression (LR): Linear baseline model.
2. Random Forest (RF): Ensemble of 300 decision trees with bootstrap aggregation.
3. MLP (ANN): Two-layer neural network (64, 32 neurons, ReLU activation).
4. XGBoost: Gradient-boosted trees with regularization.
5. Gradient Boosting (GBM): Scikit-learn's gradient boosting with 300 estimators and depth-4 trees.

V. Results

A. Model Comparison

Table 2 summarizes performance on the held-out test set with IoT features and SMOTE.

Table 2. Model performance summary (IoT feature, SMOTE).

Fig. 1. shows model comparison across accuracy.

Model	CV Acc.	Test Acc.	F1	AUC
Logistic Reg.	0.769± 0.064	0.788	0.800	0.874
Random Forest	0.835± 0.020	0.818	0.838	0.923
MLP(ANN)	0.569± 0.059	0.742	0.761	0.830
XGBoost	0.815± 0.043	0.788	0.806	0.900
Grad. Boosting	0.846± 0.040	0.833	0.849	0.911

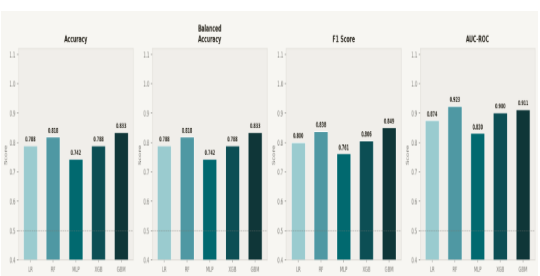


Fig. 1. Model comparison across Accuracy

Random Forest achieved the best AUC (0.923) with the lowest CV standard deviation (±0.020), indicating strong stability. MLP showed the lowest performance (AUC=0.830) and high CV variance, suggesting that n=271 is insufficient for reliable neural network training. The confusion matrix in Fig. 2, reveals 31 true positives and only 2 false negatives. The false positive rate (10 safe wells classified as unsafe) is acceptable: a false positive leads to unnecessary follow-up testing, not a health risk.

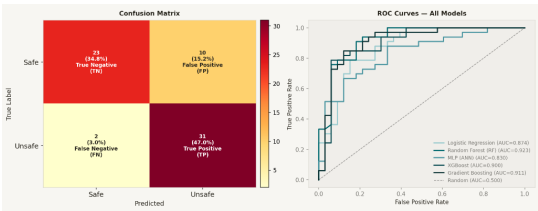


Fig. 2. Random Forest confusion matrix and ROC curves for all models.

B. Per-Class Performance

As shown in Fig. 3, Gradient Boosting achieved the highest unsafe-class recall (0.94) and F1 (0.85). Random Forest also achieved unsafe recall of 0.94 and safe recall of 0.70. Both tree ensemble methods are well-suited for this application.



Fig. 3. Per-class Precision, Recall, and F1 Score for all models.

C. Scenario Comparison

This is the central result of the paper (Fig. 4). With 63 laboratory features, RF achieves AUC=0.983 and F1=0.930. With only 6 IoT features, RF achieves AUC=0.923 and F1=0.838, retaining 93.8% of full-lab AUC performance. Removing Fe/Mn (sensor-only, 4 features) drops AUC to 0.847. The USD 0.50 per-test cost of Fe and Mn strips is therefore justified.

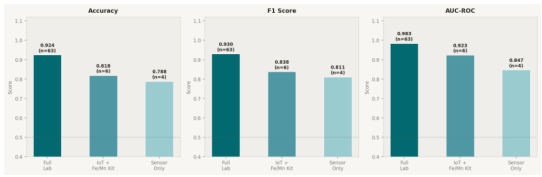


Fig. 4. Scenario comparison

D. SHAP Feature Importance

SHAP analysis (Fig. 5) reveals that Fe is the single most important predictor (mean $|\phi i|$ = 0.142), followed by pH(0.100), well depth (0.079), and temperature (0.076). High Fe values push predictions toward "unsafe," consistent with the reductive dissolution mechanism. High pH values associate with safe predictions. Shallow well depths push toward unsafe predictions. These findings align with established geochemical understanding [6].

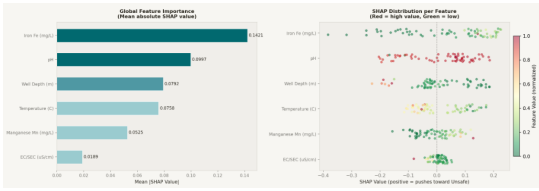


Fig. 5. SHAP feature importance for Random Forest.

E. Threshold Sensitivity

As shown in Fig. 6, at the WHO guideline of 10 µg/L (158 unsafe wells), RF achieves AUC = 0.977 and F1 = 0.939.At Bangladesh's 50 µg/L standard (108 unsafe wells),AUC = 0.923. At 100 µg/L (76 unsafe wells), AUC = 0.945. Tightening the standard toward the WHO guideline would improve model performance.

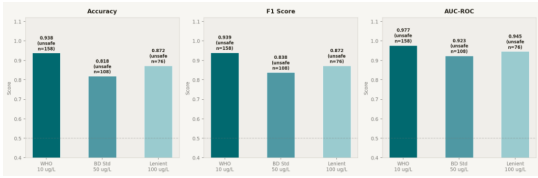


Fig. 6. Threshold sensitivity analysis

F. Cross-Dataset Generalization

As shown in Fig. 7, training on the Special Study (n = 271) and testing on the National Survey (n = 3,534) yields AUC = 0.756 and F1 = 0.493. The

decline is primarily due to feature mismatch (only Fe, Mn, and well depth are common) and spatial distributional shift. For a screening application where positives trigger follow-up testing, the 28% false positive rate is acceptable.

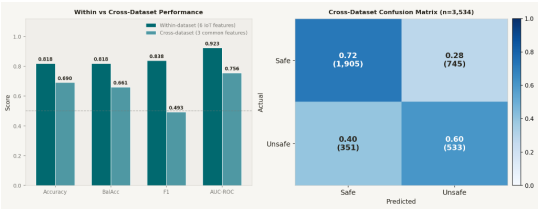


Fig. 7. Cross-dataset generalization.

VI. Conclusion

This study demonstrates that a machine learning classifier trained on six low-cost, field-measurable parameters – pH, electrical conductivity, temperature, well depth, iron, and manganese – can predict groundwater arsenic safety at the Bangladesh national standard threshold with AUC=0.923 and F1=0.838. This represents 93.8% of the discriminative performance achievable with a full 63-variable laboratory analysis at a hardware cost of approximately USD 20 and USD 0.50 per test . Random Forest was the best-performing model, and SHAP analysis confirmed that predictions are grounded in geochemically sound feature relationships.

References

[1] World Health Organization, "Arsenic," WHO Fact Sheets.

[2] A. H. Smith, E. O. Lingas, and M. Rahman, "Contamination of drinking-water by arsenic in Bangladesh: a public health emergency," Bull. World Health Organ., vol. 78, no. 9, pp. 1093-1103, 2000.

[3] International Agency for Research on Cancer (IARC), "Arsenic, Metals, Fibres and Dusts," in IARC Monographs on the Evaluation of Carcinogenic Risks to Humans, vol. 100C. Lyon, France: IARC, 2012.

[4] M. P. Murray and S. R. Rahman, "Groundwater arsenic and education attainment in Bangladesh," J. Health Popul. Nutr., vol. 33, p. 20, 2015.

[5] M. J. Morshed et al., "IoT based real-time water quality monitoring system in water treatment plants (WTPs)," Heliyon, vol. 10, 2024

[6] R. Nickson, J. McArthur, W. Burgess, K. M. Ahmed, P. Ravenscroft, and M. Rahman, "Arsenic poisoning of Bangladesh groundwater," Nature, vol. 395, p. 338, 1998.

[7] C. F. Harvey et al., "Arsenic mobility and groundwater extraction in Bangladesh," Science, vol. 298, pp. 1602-1606, 2002.

[8] F. Ullah et al., "Machine learning prediction of groundwater arsenic contamination using water quality parameters in the coastal region of Bangladesh," Environ. Geochem. Health, 2025

[9] British Geological Survey (BGS), "Arsenic contamination of groundwater in Bangladesh," 2001.